Week 2

Project Evaluation, Business Case & Feasibility

Chapter 2 — Project Evaluation and Programme Management

Central question: Should this software project be approved and formally started?

ABC Retail

Online Shopping System

Budget request
900,000 EGP

Required delivery
6 months

Approve? Revise? Reject?

Where Are We in the Project Management Cycle?

The whole course follows the management cycle introduced in Week 1.



Today we are still at the beginning of the cycle: INITIATION

Initiation asks whether the project should exist and who authorizes it. Planning starts after the project is selected.

How the Course Follows the Cycle

Week 2 goes deeper into initiation and authorization.

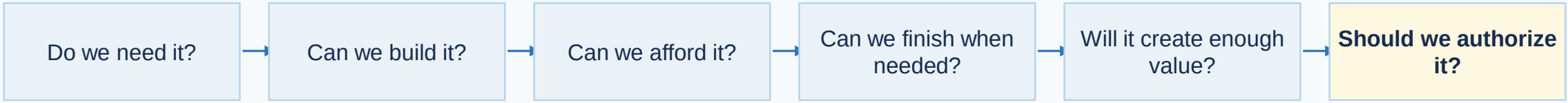
Cycle stage	Course weeks	Main course focus
Initiation	Week 1 overview + Week 2 detailed	Business need, feasibility, business case, charter, authorization
Planning	Weeks 3–7	Scope, approach, estimation, scheduling, resources, risk
Execution	Weeks 10–11	Agile execution, team coordination, sprint/release management
Monitoring & Control	Weeks 11–12	Performance tracking, EVM, dashboards, changes
Closing	Week 13	Acceptance, handover, lessons learned, project closeout

Week 2 question: Should this software project be approved and formally started?

Week 2 Management Question

Chapter 2 is about evaluating whether a proposed project should proceed.

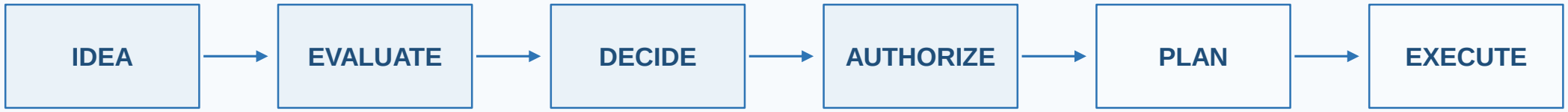
Would you spend 900,000 EGP of company money on this software project?



A project can be technically possible and still be a poor business investment.

A Software Idea Is Not Yet a Project

A manager’s request must pass through evaluation before authorization.



CEO: “Students like AI. Let’s build an AI chatbot.”

Project manager: “What problem are we solving, and why now?”

Chapter 2 Roadmap

Project Evaluation and Programme Management — the topics we will cover today.

Business Case
Why invest?

Project Selection
Which project deserves resources?

Feasibility
Can it work in this context?

Cost–Benefit
Is the value worth the cost?

Risk Evaluation
What uncertainty affects selection?

Programme & Portfolio
How does it fit wider strategy?

Business Need: Start With the Problem

A good business case begins with a business problem or opportunity.

Weak statement
“We need a new website.”

Better statement
“ABC Retail lacks an integrated digital sales channel, causing manual order handling, duplicated orders and poor order visibility.”

Interactive Check: Problem or Solution?

Classify each statement before we continue.

Statement	Problem or solution?	Why?
We need a mobile application.		
Customers wait 20 minutes to place orders.		
We need an AI chatbot.		
Staff manually enter the same data three times.		

High-Level Objectives: Make Success Discussable

SMART objectives support authorization and later evaluation.

Weak objective
“Build the best online shopping system.”

→

Improved objective
“Develop and deploy the first operational version within 6 months supporting registration, catalog, cart, payment and order tracking.”

- **Specific**
- **Measurable**
- **Achievable**
- **Relevant**
- **Time-bound**

SMART objectives are supporting Week 2 content; detailed scope is developed in Week 3.

What Is a Business Case?

The business case justifies the investment before detailed planning.

Business Case = WHY?

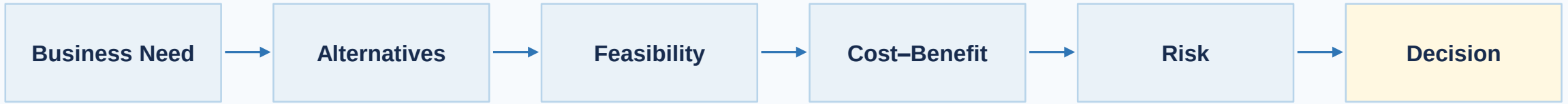
Why should the organization commit money, people and time to this software project?

Business problem / opportunity
Proposed solution and alternatives
Expected benefits
Preliminary costs
Feasibility findings
Major risks
Recommendation

The business case supports the decision; it is not yet a detailed project plan.

From Business Case to Management Decision

The output is not always a simple “yes”.



APPROVE
Proceed

MODIFY
Change scope/approach

DEFER
Wait for conditions

REJECT
Do not proceed

Interactive: which decision seems likely for ABC Retail after the first evidence.

Project Selection: Organizations Cannot Do Everything

Chapter 2 expands evaluation from one project to a portfolio of possible projects.

Limited money

Limited people

Limited attention

Portfolio thinking asks: Which projects best support strategy and provide value under constraints?

A good project may still be delayed if a better strategic project needs the same resources.

Interactive Portfolio Example

ABC Retail has only 1.2 million EGP available this year.

Proposal	Cost	Expected benefit	Strategic importance
Online Shopping System	700k	High	High
HR Mobile App	350k	Medium	Low
Warehouse Automation	800k	Very high	Very high

Question: Which combination should management fund — and what criteria should guide the decision?

There is no single calculation-only answer; students must defend a selection approach.

Project, Programme, Portfolio

Chapter 2 connects project evaluation to programme management.

Concept	Meaning	Example
Project	A temporary undertaking with a defined output	Build the Online Shopping System
Programme	A coordinated group of related projects delivering broader benefits	Digital Commerce Programme
Portfolio	A collection of projects/programmes selected to support strategy	All IT transformation investments

A programme focuses on coordinated benefits, not merely a bigger project.

Programme Management Example

Related projects can be coordinated to deliver strategic business benefits.

DIGITAL COMMERCE PROGRAMME

- **Online Store Project**
- **Mobile App Project**
- **Warehouse Integration**
- **CRM Upgrade**

Programme benefit: better digital customer experience and coordinated online sales capability

Use this to show why an Online Shopping project may be evaluated as part of wider transformation.

Feasibility: Can This Project Work in This Context?

Feasibility is broader than “Can programmers write the code?”

Technical
Can we build/integrate it?

Economic
Is value worth cost?

Operational
Will the business use it?

Schedule
Can it finish when needed?

Organizational
Do we have support/resources?

Legal / Compliance
Can we operate it responsibly?

Technical feasibility alone is never enough for authorization.

Technical Feasibility: ABC Retail Example

The project is possible, but the technical risk may be significant.

Required features
- Web application
- Online payment
- Real-time inventory
- Courier integration
- Admin dashboard

Current capability
- 2 Java developers
- 1 business analyst
- No secure-payment specialist
- No UI/UX contractor assigned

Prompt: Does missing expertise mean the project must be rejected? Or should conditions be added?

Operational Feasibility: The Perfect System Nobody Uses

A working system can fail if business processes and users do not adopt it.

Scenario
A CRM system is delivered and works correctly.

But sales staff continue to keep customer data in personal Excel files.

Operational questions
- Is workflow changing?
- Are users trained?
- Is management enforcing use?
- Does the system fit the real process?

Ask: Did the software project succeed if nobody uses the system properly?

Schedule Feasibility: “CEO Mathematics”

Adding people does not always reduce duration proportionally.

CEO: “If one developer needs 12 months, let’s assign 12 developers and finish in 1 month.”

Reality: software work has dependencies, learning, integration, testing and communication overhead.

Preliminary phase	Duration
Requirements + Design + Development + Testing + Deployment	34 weeks total if simply added
Management request	6 months

Tell students: Week 5 will teach formal scheduling; here we identify schedule feasibility risk.

Economic Feasibility: Value vs Investment

The same technology can be a good or bad investment depending on the numbers and context.

Case A
Cost = 1,000,000 EGP
Annual saving = 70,000 EGP
Useful life = 3 years

Likely weak economics

Case B
Cost = 1,000,000 EGP
Annual saving = 600,000 EGP
Useful life = 3 years

Potentially attractive

Same idea, different economic case.

Costs and Benefits: What Goes Into Evaluation?

Chapter 2 cost–benefit evaluation requires identifying both sides clearly.

Costs	Examples
One-time	Development, setup, data migration, training
Recurring	Hosting, licenses, maintenance, support
Hidden/overlooked	Change management, integration, user training

Benefits	Examples
Tangible	More sales, fewer errors, lower labor cost
Intangible	Customer satisfaction, reputation, better information
Strategic	Digital capability, market reach, future growth

Intangible does not mean unimportant; it means harder to quantify.

Worked Example: Simple Cash-Flow View

Start with a clear project-evaluation spreadsheet.

Item	Amount
Initial project cost	700,000 EGP
Expected annual contribution / savings	450,000 EGP
Annual cloud/support/maintenance	120,000 EGP
Net annual benefit	330,000 EGP

Cost–Benefit Technique: Payback Period

How long before the investment is recovered?

Payback Period = Initial Investment ÷ Net Annual Benefit

700,000 ÷ 330,000 ≈ 2.12 years

Important: Shorter payback is useful evidence — but not the whole decision.

Cost–Benefit Technique: Simplified ROI

Return on investment summarizes value relative to cost.

ROI = (Total Benefit − Total Cost) ÷ Total Cost × 100

3-year evaluation item	Calculation	Result
Gross benefits	450,000 × 3	1,350,000
Total costs	700,000 + (120,000 × 3)	1,060,000
ROI	(1,350,000 − 1,060,000) ÷ 1,060,000	≈ 27.36%

Time Value of Money: NPV Concept

Chapter 2 often introduces that money today is not the same as money later.



NPV discounts future cash flows to today’s value before comparing costs and benefits.

Interactive: Choose Using More Than One Number

Shortest payback is not automatically the best project.

Project	Cost	Payback	Risk	Strategic importance
A: Online discounts engine	700k	2 years	High cybersecurity risk	Low
B: Warehouse integration	900k	3 years	Low technical risk	High

Question: Which project would you recommend — and what evidence supports your decision?

Risk Evaluation During Project Selection

Chapter 2 risk evaluation asks how uncertainty affects the approval decision.

What could prevent benefits from being realized?
What could increase costs or delay delivery?
What assumptions could prove false?
What risks are unacceptable at the selection stage?
What conditions should be added before approval?

ABC Retail risk examples
- Payment API delay
- Missing security expertise
- Warehouse process not ready
- Unrealistic 6-month target
- Low customer adoption

This is initial risk evaluation. Full risk register comes in Week 7.

Risk Can Change the Selection Decision

Two projects with similar benefits may not be equally attractive.

Project	Expected benefit	Selection risk	Possible management response
Project A	High	High supplier dependency	Approve only with vendor SLA
Project B	Medium	Low risk and strategic fit	Approve as planned
Project C	Very high	Unproven technology	Pilot first, then decide

Question: Which risk response belongs in the business case recommendation?

Feasibility Is Not Simply Yes / No

Evaluation should create options for decision makers.

Proceed as proposed

Proceed with conditions

Reduce scope

Delay until ready

Reject or replace solution

Example: Approve Phase 1 as responsive web; move mobile app and advanced payment options to Phase 2.

Feasibility Matrix for ABC Retail

Ratings are useful only when supported by evidence and action.

Area	Rating	Concern	Recommended action
Technical	3/5	Payment expertise missing	Hire/contract specialist
Economic	4/5	Positive value estimate	Validate benefits with finance
Operational	3/5	Warehouse readiness medium	Improve inventory process
Schedule	3/5	Six-month deadline tight	Validate detailed schedule in Week 5
Organizational	4/5	Sponsor support high	Maintain engagement

Assumptions, Constraints and Risks

These are high-level inputs to authorization and later planning.

Concept	Meaning	ABC Retail example
Assumption	Believed true for planning	Payment API will be available next month
Constraint	A boundary that limits choices	Budget cannot exceed 900,000 EGP
Risk	Uncertain event that may affect objectives	Payment provider may delay integration
Issue	A risk/event that has already occurred	API documentation is actually delayed

AUTHORIZATION

Project Charter and Sponsor Authorization

After selection, the charter formally initiates the project.

Project Charter includes
- Purpose and business need
- High-level objectives
- High-level scope
- Major deliverables
- Initial budget and target date
- Major risks
- Project Manager
- Sponsor approval

Sponsor Authorization means
- **Project is officially approved**
- **PM receives authority**
- **Resources can be committed**
- **Planning can proceed**

AUTHORIZATION

Business Case vs Project Charter

Students must not confuse the decision document with the authorization document.

Business Case	Project Charter
Answers: Why should we invest?	Answers: What project is being authorized?
Compares alternatives and benefits	Formally establishes the project
Supports selection decision	Gives authority to the Project Manager
Before approval	After approval / at initiation

Memory aid: Business Case = WHY? | Project Charter = GO!

Step Wise Preview: Planning Is More Than a Gantt Chart

We preview the framework now; we study it formally in Week 3.

Should we do the project?

What are we trying to achieve?

What products and activities are required?

How much effort is needed?

What can go wrong?

Who will do the work?

Is the plan realistic?

STEP WISE PREVIEW

The Step Wise Planning Framework

A summarized view of the model used in the textbook.

Stage	Step Wise steps	Question
Project selection	Step 0 — Select Project	Should we undertake this project?
Project definition	Steps 1–3	What objectives, scope, infrastructure and characteristics define it?
Project planning	Steps 4–8	What products, activities, effort, risks and resources are required?
Execution & refinement	Steps 9–10	How do we execute and update detailed plans?

Do not ask students to memorize all steps today; focus on the logic.

STEP WISE PREVIEW

Where Are We in Step Wise?

Week 2 mainly corresponds to Step 0 — Select Project.

STEP 0 — SELECT PROJECT

- **Business need**
- **Alternatives**
- **Feasibility**
- **Costs and benefits**
- **Initial risks**
- **Business case**
- **Selection decision**

If approved:
Project Charter
↓
Sponsor Authorization
↓
Week 3: Step 1
Scope & Objectives

This is the clean bridge from Chapter 2 to Chapter 3.

APPLICATION

Final Interactive Case: Authorization Board

Return to the opening question with evidence.

Evidence	ABC Retail status
Initial project cost	700,000 EGP; max authorized budget 900,000 EGP
Annual net benefit	330,000 EGP; payback ≈ 2.12 years
Technical concern	Payment-security expertise missing
Operational concern	Warehouse readiness medium
Schedule concern	Six-month target is tight

Decision options: Approve • Approve with conditions • Revise • Defer • Reject

RECAP

Chapter 2 Takeaways

What students should remember after this lecture.

A business case justifies the investment before detailed planning.
Project selection compares value, feasibility, risk and strategic fit.
Feasibility has multiple dimensions: technical, economic, operational, schedule and organizational.
Cost–benefit techniques support decisions, but do not replace management judgment.
Programmes coordinate related projects to realize broader benefits.
Authorization is formalized through the project charter and sponsor approval.

RECAP

Transition to Week 3

Once the project is authorized, the question changes.

Week 2
Should we do this project?

Evaluation
Business Case
Authorization

Week 3
What exactly are we committing to deliver?

Scope
Requirements
WBS
Step Wise planning

RECAP

Quick Recap Quiz

Use these questions in the final five minutes.

Question	Expected answer
Which document answers “Why should we invest?”	Business Case
Which document formally authorizes the project?	Project Charter
Can a technically feasible project still be rejected?	Yes — economics, risks or strategy may be weak
What does payback period measure?	Time needed to recover the investment
What Step Wise step are we mainly doing in Week 2?	Step 0 — Select Project

Final question: Would you approve ABC Retail as proposed, or approve with conditions?